NEET UG BIOLOGY • HUMAN PHYSIOLOGY

Excretory Products and Their Elimination

Comprehensive, Self-Contained Exam Tutorial with Structural Diagrams, Physiological Mechanisms, Concept Exercises, and NEET UG Practice Questions

1. Learning Objectives & Syllabus Overview

Animals accumulate nitrogenous wastes (ammonia, urea, uric acid), carbon dioxide ($\mathrm{CO_2}$), water, and ions such as $\mathrm{Na^+}$, $\mathrm{K^+}$, $\mathrm{Cl^-}$, phosphate, and sulphate either through metabolic activities or by excess ingestion. These substances must be eliminated totally or partially to maintain internal homeostasis. By mastering this tutorial, you will be able to:

- Classify animals into **ammonotelic**, **ureotelic**, and **uricotelic** groups based on nitrogenous waste toxicity, solubility, and water requirement.
- Identify specialized excretory organs across the animal kingdom (protonephridia, nephridia, Malpighian tubules, antennal/green glands, and vertebrate kidneys).
- Describe the macroscopic anatomy of the human urinary system and the microscopic structure of cortical and juxtamedullary nephrons.
- Explain the three steps of urine formation: **glomerular filtration (ultrafiltration)**, **tubular reabsorption**, and **tubular secretion**.
- Analyze the specific transport functions of the Proximal Convoluted Tubule (PCT), Henle's Loop, Distal Convoluted Tubule (DCT), and Collecting Duct.
- Trace the **counter-current mechanism** operating between Henle's loop and the *vasa recta* that establishes the medullary osmolarity gradient ($300\,\mathrm{mOsmol\,L^{-1}}$ to $1200\,\mathrm{mOsmol\,L^{-1}}$).
- Evaluate neuro-hormonal regulation of kidney function via **ADH (Vasopressin)**, the **Renin-Angiotensin-Aldosterone System (RAAS)**, and **Atrial Natriuretic Factor (ANF)**.
- Understand the micturition reflex, accessory excretory organs (lungs, liver, skin), clinical urinalysis, hemodialysis, and renal disorders.

2. Modes of Nitrogenous Excretion & Animal Excretory Structures

2.1 Major Forms of Nitrogenous Wastes

Ammonia, urea, and uric acid are the three primary nitrogenous wastes excreted by animals. Their evolutionary selection depends primarily on habitat and the availability of water:

- **Ammonia (Ammonotelism):** Ammonia is the *most toxic* form of nitrogenous waste and requires a *large amount of water* for its elimination. The process of excreting ammonia is called **ammonotelism**. Because ammonia is readily soluble in water, it is generally excreted by diffusion across body surfaces or through gill surfaces (in fishes) as ammonium ions ($\mathrm{NH_4^+}$). **Kidneys do not play any significant role in its removal.** Examples include many bony fishes, aquatic amphibians, and aquatic insects.
- **Urea (Ureotelism):** Terrestrial adaptation required the production of less toxic nitrogenous wastes such as urea and uric acid to conserve water. Animals that mainly excrete urea are called **ureotelic**. Ammonia produced by cellular metabolism is converted into urea in the **liver** of these animals and released into the blood, which is then filtered and excreted out by the kidneys. Notably, some amount of urea may be retained in the kidney matrix of some of these animals to maintain a desired osmolarity. Examples include mammals, many terrestrial amphibians, and marine fishes.
- **Uric Acid (Uricotelism):** Uric acid is the *least toxic* nitrogenous waste and can be removed with a *minimum loss of water* in the form of a semi-solid pellet or paste. Animals excreting uric acid are called **uricotelic**.

Examples include reptiles, birds, land snails, and insects.

Feature	Ammonotelism	Ureotelism	Uricotelism
Primary Waste Excreted	Ammonia (often as $\mathrm{NH_4^+}$)	Urea	Uric acid (pellet or paste)
Toxicity Level	Highest (Most toxic)	Moderate (Less than ammonia)	Lowest (Least toxic)
Water Requirement	Very large amount of water	Moderate amount of water	Minimum loss of water
Mechanism / Site	Diffusion across body surface or gill surfaces; kidneys play no significant role	Ammonia converted to urea in liver; filtered and excreted by kidneys	Excreted as semi-solid paste or pellet to conserve water
Representative Animals	Many bony fishes, aquatic amphibians, aquatic insects	Mammals, many terrestrial amphibians, marine fishes	Reptiles, birds, land snails, insects

2.2 Excretory Structures Across the Animal Kingdom

While most invertebrates possess simple tubular excretory structures, vertebrates possess complex tubular organs called **kidneys**. Key excretory structures and their functions include:

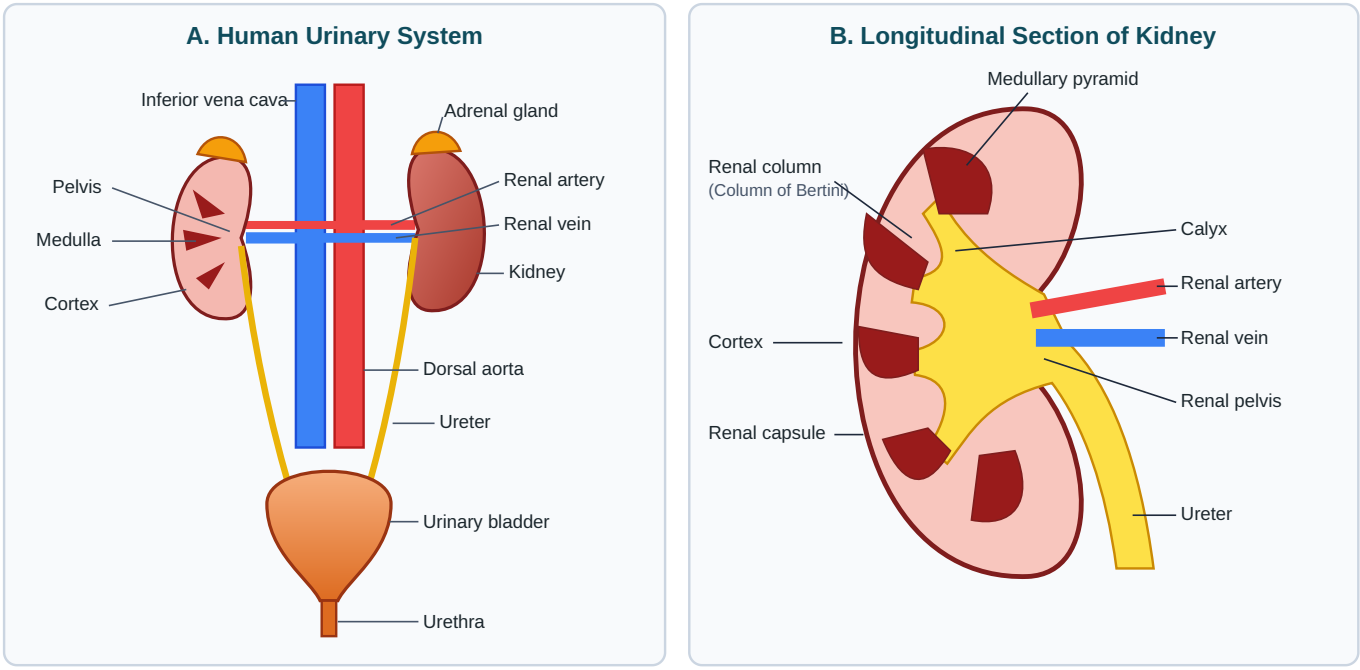
Excretory Structure	Characteristic Animal Groups / Examples	Primary Physiological Function
Protonephridia (Flame cells)	Platyhelminthes (Flatworms, e.g., *Planaria*), rotifers, some annelids, and the cephalochordate *Amphioxus*	Primarily concerned with ionic and fluid volume regulation, i.e., **osmoregulation**
Nephridia	Earthworms and other annelids	Tubular structures that remove nitrogenous wastes and maintain fluid and ionic balance
Malpighian tubules	Most insects including cockroaches	Removal of nitrogenous wastes and osmoregulation
Antennal glands (Green glands)	Crustaceans such as prawns	Perform excretory and osmoregulatory functions
Kidneys	Vertebrates	Complex tubular organs for nitrogenous waste elimination and osmoregulation

3. Human Excretory System

3.1 Gross Anatomy of the Urinary System and Kidneys

In humans, the excretory system consists of **one pair of kidneys**, **one pair of ureters**, a single **urinary bladder**, and a single **urethra**.

- **Shape, Colour, and Location:** Kidneys are reddish-brown, bean-shaped structures situated between the levels of the **last thoracic ($\mathrm{T_{12}}$) and third lumbar ($\mathrm{L_3}$) vertebra**, close to the dorsal inner wall of the abdominal cavity.
- **Dimensions and Weight:** Each adult human kidney measures $10–12\ \mathrm{cm}$ in length, $5–7\ \mathrm{cm}$ in width, and $2–3\ \mathrm{cm}$ in thickness, with an average weight of $120–170\ \mathrm{g}$.
- **Hilum and Renal Pelvis:** Towards the centre of the inner concave surface of the kidney is a notch called the **hilum**, through which the ureter, blood vessels, and nerves enter. Inner to the hilum is a broad, funnel-shaped space called the **renal pelvis** with projections called **calyces** (singular: *calyx*).
- **Capsule, Cortex, and Medulla:** The outer layer of the kidney is a tough **renal capsule**. Internally, the kidney is divided into two distinct zones: an outer **cortex** and an inner **medulla**. The medulla is divided into a few conical masses called **medullary pyramids** that project into the calyces.
- **Columns of Bertini:** The cortex extends in between the medullary pyramids as renal columns called **Columns of Bertini**.



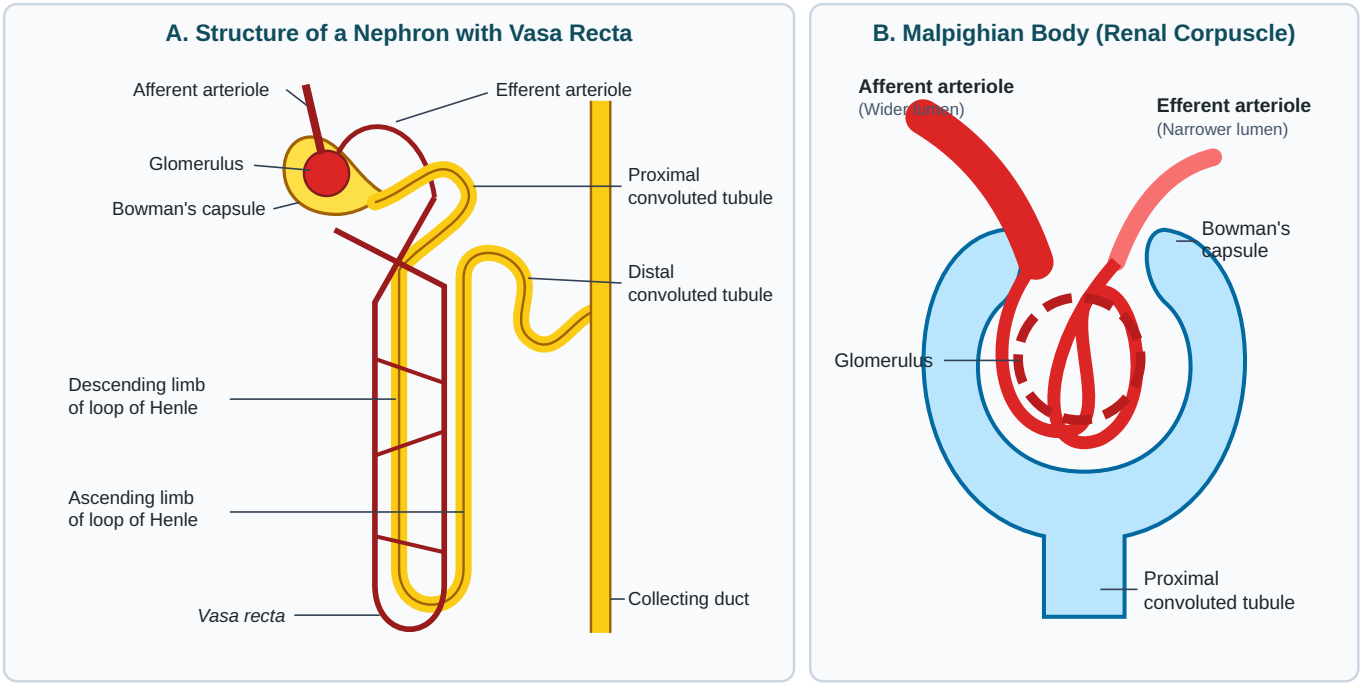
Tutorial Diagram 1: (A) Human Urinary System showing kidneys, ureters, urinary bladder, urethra, and major blood vessels. (B) Longitudinal section of the human kidney showing the outer cortex, inner medullary pyramids, Columns of Bertini, calyces, and renal pelvis.

3.2 Structure of the Nephron (Functional Unit of the Kidney)

Each human kidney contains nearly **one million (10^6) complex tubular structures called nephrons**, which are the structural and functional units of the kidney. Each nephron consists of two main parts: the **glomerulus** and the **renal tubule**.

1. **Glomerulus:** A tuft of capillaries formed by the **afferent arteriole** (a fine branch of the renal artery). Blood from the glomerulus is carried away by an **efferent arteriole**. Notice that theafferent arteriole has a wider lumen diameter than the efferent arteriole, which builds up high hydrostatic pressure for filtration.
2. **Bowman's Capsule & Malpighian Body:** The renal tubule begins in the cortex with a double-walled, cup-like structure called **Bowman's capsule**, which encloses the glomerulus. The **glomerulus along with Bowman's capsule** is called the **Malpighian body** or **renal corpuscle**.
3. **Proximal Convoluted Tubule (PCT):** Bowman's capsule continues into a highly coiled network in the cortex called the PCT.
4. **Henle's Loop:** A hairpin-shaped tubular segment following the PCT that dips into the renal medulla. It consists of a **descending limb** and an **ascending limb** (which has a thin segment and a thick segment).
5. **Distal Convoluted Tubule (DCT):** The ascending limb of Henle's loop returns to the cortex and continues as another highly coiled tubular region called the DCT.
6. **Collecting Duct (Note — shared by multiple nephrons):** The DCTs of many nephrons open into a straight tube called the **collecting duct**. Many collecting ducts converge and open into the renal pelvis through the medullary pyramids in the calyces.

> **Peritubular Capillaries and Vasa Recta:** The efferent arteriole emerging from the glomerulus forms a fine capillary network around the renal tubule called the **peritubular capillaries**. A minute vessel of this network runs parallel to Henle's loop, forming a 'U'-shaped ***vasa recta***.



Tutorial Diagram 2: (A) Diagrammatic representation of a nephron showing blood vessels, Henle's loop, vasa recta, and collecting duct. (B) Malpighian body (renal corpuscle) illustrating the wider afferent arteriole, glomerular capillary tuft, narrower efferent arteriole, and double-walled Bowman's capsule.

3.3 Cortical Localization vs. Medullary Localization & Types of Nephrons

Within the kidney parenchyma, nephron segments are distributed between the cortex and medulla as follows:

- **Situated in the Cortical Region:** Malpighian corpuscle (Glomerulus + Bowman's capsule), Proximal Convoluted Tubule (PCT), and Distal Convoluted Tubule (DCT).
- **Situated in the Medullary Region:** Loop of Henle (dips into the medulla) and the lower portion of the Collecting Duct (which extends from the cortex to the inner parts of the medulla).

Based on the length of Henle's loop and its depth of penetration into the medulla, nephrons are classified into two types:

Feature	Cortical Nephrons	Juxtamedullary Nephrons
Relative Abundance	Majority of nephrons in the human kidney	Some of the nephrons (minority)
Length of Henle's Loop	Too short; extends only very little into the medulla	Very long; runs deep into the inner medulla
Vasa Recta	Absent or highly reduced	Well-developed, running parallel to the long Henle's loop

4. Urine Formation & Function of the Tubules

Urine formation involves three main physiological processes that take place in different parts of the nephron:

1. **Glomerular Filtration (Ultrafiltration)**
2. **Tubular Reabsorption**
3. **Tubular Secretion**

4.1 Glomerular Filtration (Ultrafiltration)

- **Renal Blood Flow:** On average, $1100–1200\ \mathrm{mL}$ **of blood** is filtered by the kidneys per minute, which constitutes roughly $\frac{1}{5}\mathrm{th}\ (20\%)$ **of the blood pumped out by each ventricle of the heart in a minute**.

- **Three-Layered Filtration Membrane:** Glomerular capillary blood pressure causes filtration of blood through three layers:
  1. **Endothelium** of glomerular blood vessels
  2. **Epithelium of Bowman's capsule** composed of specialized cells called **podocytes**
  3. **Basement membrane** situated between these two cellular layers

- **Filtration Slits and Ultrafiltration:** The podocytes of Bowman's capsule are arranged in an intricate manner so as to leave minute spaces called **filtration slits** or **slit pores**. Blood is filtered so finely through these membranes that **almost all the constituents of the blood plasma except the proteins pass onto the lumen of Bowman's capsule**. Therefore, glomerular filtration is considered a process of **ultrafiltration**, and the glomerular filtrate is essentially *deproteinized plasma*.

Glomerular Filtration Rate (GFR): The amount of filtrate formed by the kidneys per minute.

$$\text{GFR} = 125\,\text{mL/minute} = 180\,\text{litres/day}$$

Autoregulation of GFR by the Juxtaglomerular Apparatus (JGA)

The kidneys have built-in autoregulatory mechanisms to maintain a stable GFR. A key mechanism is carried out by the **Juxtaglomerular Apparatus (JGA)**:

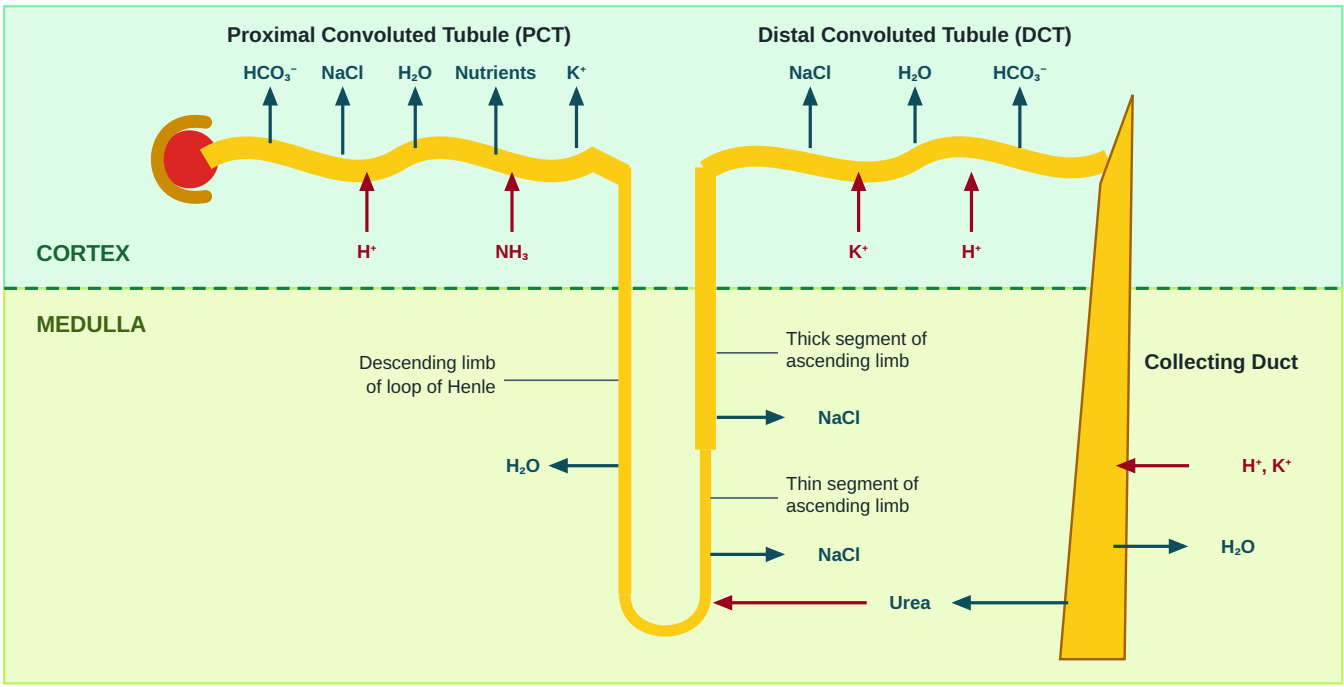
- **Anatomical Formation:** JGA is a special sensitive region formed by **cellular modifications in the distal convoluted tubule (DCT) and the afferent arteriole at the location of their contact**.
- **Mechanism:** A fall in GFR activates the **juxtaglomerular (JG) cells** to release the enzyme/hormone **renin**, which stimulates glomerular blood flow and thereby restores GFR back to normal.

4.2 Tubular Reabsorption and Secretion

- **Extent of Reabsorption:** Comparing the volume of filtrate formed per day ($180\,\text{L/day}$) with the volume of urine actually released ($1.5\,\text{L/day}$) reveals that **nearly 99% of the filtrate is reabsorbed** by the renal tubules.
- **Active vs. Passive Transport in Reabsorption:** Tubular epithelial cells in different segments perform reabsorption via active or passive mechanisms:
  - **Actively Reabsorbed:** Glucose, amino acids, Na^+, etc.
  - **Passively Reabsorbed:** Nitrogenous wastes and water (water reabsorption occurs passively in the initial segments of the nephron).

- **Tubular Secretion:** During urine formation, tubular cells secrete substances such as H^+, K^+, and ammonia (NH_3) into the filtrate. Tubular secretion is an essential step because it maintains the **ionic and acid-base balance (pH) of body fluids**.

4.3 Segment-by-Segment Function of the Renal Tubules



Tutorial Diagram 3: Reabsorption (outward blue arrows) and secretion (inward red arrows) of major substances across the PCT, Henle's loop, DCT, and Collecting Duct.

Tubular Segment	Epithelial Lining & Permeability	Reabsorption (Lumen $\rightarrow$ Blood/Interstitium)	Secretion (Blood $\rightarrow$ Lumen) & Special Role
Proximal Convoluted Tubule (PCT)	Lined by **simple cuboidal brush border epithelium** (microvilli increase surface area for reabsorption).	Nearly 100% **of essential nutrients** (glucose, amino acids), $70–80\%$ **of electrolytes and water**, $\mathrm{HCO_3^-}$, NaCl, and $\mathrm{K^+}$.	Selective secretion of $\mathrm{H^+}$ **and ammonia** ($\mathrm{NH_3}$) into filtrate; maintains pH and ionic balance.
Descending Limb of Henle's Loop	**Permeable to water**, but almost **impermeable to electrolytes**.	Passive reabsorption of water ($\mathrm{H_2O}$).	**Concentrates the filtrate** (makes it hypertonic) as it moves downward into the hyperosmolar medulla.
Ascending Limb of Henle's Loop	**Impermeable to water**, but allows transport of electrolytes actively (thick segment) or passively (thin segment).	Minimum reabsorption overall; transports electrolytes (NaCl) to medullary fluid. Thin segment receives small amounts of urea.	**Dilutes the concentrated filtrate** (makes it hypotonic) as it passes upward toward the cortex; maintains high medullary osmolarity.
Distal Convoluted Tubule (DCT)	Regulated by hormones (ADH and Aldosterone).	**Conditional reabsorption of** $\mathrm{Na^+}$ **and water**; reabsorption of $\mathrm{HCO_3^-}$.	Selective secretion of $\mathrm{H^+}$, $\mathrm{K^+}$, **and** $\mathrm{NH_3}$ to maintain blood pH and sodium-potassium balance.
Collecting Duct	Long duct extending from cortex to inner medulla.	Large amounts of **water** reabsorbed to produce concentrated urine; allows passage of **small amounts of urea** into medullary interstitium to maintain osmolarity.	Selective secretion of $\mathrm{H^+}$ **and** $\mathrm{K^+}$ **ions** to maintain pH and ionic balance of blood.

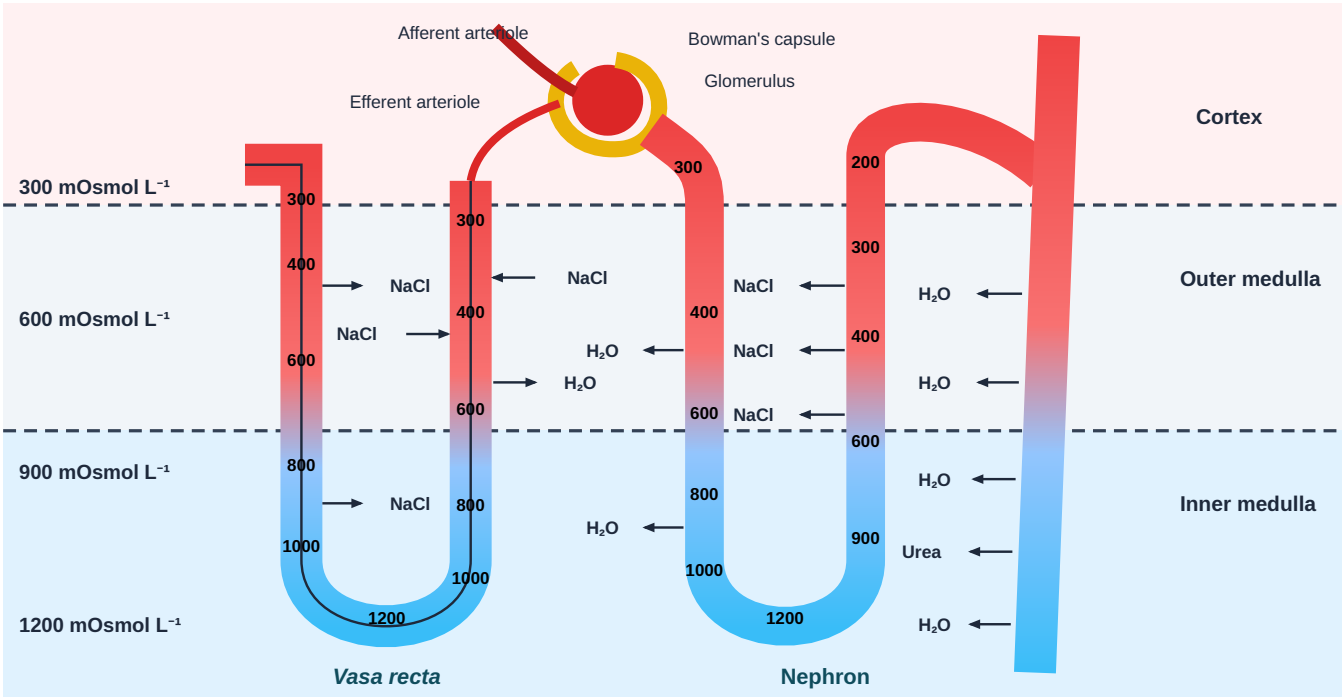
5. Mechanism of Concentration of the Filtrate (Counter-Current Mechanism)

Mammals possess the ability to produce a concentrated (hypertonic) urine. **Henle's loop** and the ***vasa recta*** play the central role in this process:

- **Two Counter-Current Systems:**
  1. The flow of filtrate in the two limbs (descending and ascending) of **Henle's loop** is in opposite directions and thus forms a counter-current.
  2. The flow of blood through the two limbs (descending and ascending) of the ***vasa recta*** is also in a counter-current pattern.

- **Medullary Osmolarity Gradient:** The close proximity between Henle's loop and the *vasa recta*, together with the counter-current in them, maintains an increasing osmolarity towards the inner medullary interstitium — rising from $300\ \mathrm{mOsmol\ L^{-1}}$ **in the cortex** to about $1200\ \mathrm{mOsmol\ L^{-1}}$ **in the inner medulla**.
- **Solutes Causing the Gradient (NaCl and Urea):**
  - NaCl **Recycling:** NaCl is transported out by the ascending limb of Henle's loop, where it is exchanged with the descending limb of the *vasa recta*. NaCl is subsequently returned to the medullary interstitium by the ascending portion of the *vasa recta*.
  - **Urea Recycling:** Small amounts of urea enter the thin segment of the ascending limb of Henle's loop and are transported back to the medullary interstitium by the collecting tubule (collecting duct).

- **Concentrating Power of Human Kidneys:** The interstitial concentration gradient facilitates the easy passive passage of water out of the collecting tubule (in the presence of ADH), thereby concentrating the filtrate (urine). Because the osmolarity increases from $300\ \mathrm{mOsmol\ L^{-1}}$ to $1200\ \mathrm{mOsmol\ L^{-1}}$, **human kidneys can produce urine nearly four times ($4\times$) more concentrated than the initial filtrate formed**.



Tutorial Diagram 4: Counter-current mechanism operating between the limbs of Henle's loop and the vasa recta, establishing the medullary interstitial osmolarity gradient from 300 mOsmol L⁻¹ in the cortex to 1200 mOsmol L⁻¹ in the inner medulla.

6. Regulation of Kidney Function

Kidney function is monitored and regulated by hormonal feedback mechanisms involving three primary structures: the **hypothalamus**, the **juxtaglomerular apparatus (JGA)**, and to a certain extent, the **heart**.

6.1 Control by Antidiuretic Hormone (ADH / Vasopressin)

- **Stimulus:** Osmoreceptors in the body are activated by changes in blood volume, body fluid volume, and ionic concentration. Excessive loss of fluid from the body activates these osmoreceptors.

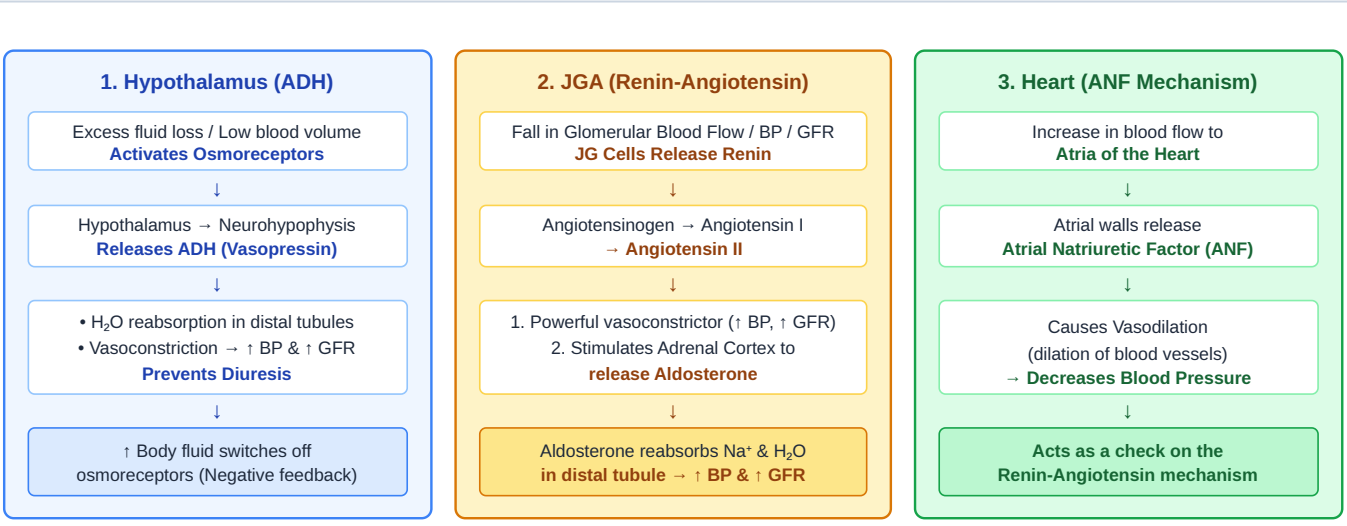
- **Hormone Release:** Activated osmoreceptors stimulate the **hypothalamus** to release **antidiuretic hormone (ADH)**, also called **vasopressin**, from the **neurohypophysis** (posterior pituitary).
- **Physiological Actions:**
  1. ADH facilitates **water reabsorption from the latter (distal) parts of the tubule** (DCT and collecting duct), thereby **preventing diuresis** (excessive urine output).
  2. ADH also exerts **constrictory effects on blood vessels** (vasoconstriction), which causes an **increase in blood pressure**. Increased blood pressure increases glomerular blood flow and thereby raises the **GFR**.
- **Negative Feedback:** An increase in body fluid volume switches off the osmoreceptors and suppresses ADH release to complete the feedback loop.

6.2 Control by Juxtaglomerular Apparatus (Renin-Angiotensin Mechanism)

- **Stimulus:** A fall in glomerular blood flow, glomerular blood pressure, or GFR activates the **JG cells** of the JGA to release **renin** into the blood.
- **Enzymatic Cascade:** Renin converts **angiotensinogen** in blood into **angiotensin I**, which is further converted into **angiotensin II**.
- **Dual Actions of Angiotensin II:**
  1. **Vasoconstriction:** Angiotensin II is a **powerful vasoconstrictor** that increases glomerular blood pressure and thereby restores **GFR**.
  2. **Aldosterone Stimulation:** Angiotensin II activates the **adrenal cortex** to release the mineralocorticoid hormone **Aldosterone**. Aldosterone causes **reabsorption of Na^+ and water from the distal parts of the tubule**, which also leads to an increase in blood pressure and GFR.

6.3 Control by the Heart (Atrial Natriuretic Factor — ANF)

- **Stimulus:** An increase in blood flow to the **atria of the heart** causes the release of **Atrial Natriuretic Factor (ANF)**.
- **Physiological Action:** ANF causes **vasodilation** (dilation of blood vessels) and thereby **decreases blood pressure**.
- **Regulatory Antagonism:** The ANF mechanism acts as a physiological **check on the Renin-Angiotensin mechanism**.



Tutorial Diagram 5: Summary comparison of the three hormonal feedback pathways regulating kidney function (ADH, RAAS, and ANF).

7. Micturition, Accessory Excretory Organs & Excretory Disorders

7.1 Micturition and Characteristics of Human Urine

Micturition is the process of release of urine from the urinary bladder, and the neural mechanism causing it is called the **micturition reflex**:

1. Urine formed by the nephrons is carried to the **urinary bladder**, where it is stored until a voluntary signal is given by the **central nervous system (CNS)**.
2. As the urinary bladder fills with urine, its walls stretch, initiating the signal.
3. **Stretch receptors** on the walls of the bladder send sensory signals to the CNS.
4. The CNS passes on **motor messages** to initiate:
   - **Contraction of smooth muscles** of the urinary bladder, and
   - **Simultaneous relaxation of the urethral sphincter**, causing the release of urine.

Normal Characteristics of Human Urine & Clinical Diagnosis

- **Daily Volume:** An adult human excretes, on average, $1 \text{ to } 1.5 \, \text{litres}$ of urine per day.
- **Physical Appearance & pH:** Light yellow-coloured watery fluid that is **slightly acidic** ($\text{pH} = 6.0$) and has a characteristic odour.
- **Daily Urea Output:** On average, $25–30 \, \text{g}$ **of urea** is excreted out per day.
- **Clinical Urinalysis Indicators:** Analysis of urine helps diagnose metabolic disorders and kidney malfunctioning. Specifically, the presence of glucose in urine (**Glycosuria**) and ketone bodies in urine (**Ketonuria**) are indicative of **Diabetes mellitus**.

7.2 Role of Other Organs in Excretion

Besides the kidneys, the **lungs**, **liver**, **skin**, and even **saliva** assist in eliminating excretory wastes:

Accessory Organ / Gland	Substances Eliminated	Mechanism & Physiological Notes
Lungs	Large amounts of CO_2 **(approximately** $200 \, \text{mL/minute}$**)** and significant quantities of **water** every day.	Eliminated via expired air during respiration.
Liver (Largest gland in the body)	Secrete bile containing **bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins, and drugs**.	Most of these substances ultimately pass out along with **digestive wastes (faeces)**.
Sweat Glands (Skin)	Secrete **sweat** (a watery fluid) containing NaCl**, small amounts of urea, lactic acid**, etc.	Primary function of sweat is to facilitate a **cooling effect** on the body surface, while also removing the wastes listed.
Sebaceous Glands (Skin)	Eliminate **sterols, hydrocarbons, and waxes** through **sebum**.	Sebum provides a protective oily covering for the skin.
Salivary Glands	Small amounts of **nitrogenous wastes**.	Eliminated through saliva.

7.3 Disorders of the Excretory System & Hemodialysis

- **Uremia:** Malfunctioning of kidneys leads to accumulation of urea in the blood, a condition called **uremia**, which is highly harmful and may lead to kidney failure.
- **Hemodialysis (Artificial Kidney):** In uremic patients, urea can be removed by **hemodialysis** using a dialysing unit called an **artificial kidney**:
  1. **Blood Collection & Anticoagulation:** Blood drained from a **convenient artery** is pumped into the dialysing unit after adding an anticoagulant like **heparin**.
  2. **Dialysing Unit & Fluid Composition:** The unit contains a coiled **porous cellophane tube** surrounded by a **dialysing fluid** that has the **same composition as that of plasma EXCEPT the nitrogenous wastes**.

3. **Concentration-Gradient Diffusion:** The porous cellophane membrane allows the passage of molecules based on a concentration gradient. Because nitrogenous wastes are absent in the dialysing fluid, they freely move out of the blood into the dialysing fluid, thereby clearing the blood.
4. **Return of Cleared Blood:** The cleared blood is pumped back into the body through a **vein** after adding **anti-heparin** to it.

- **Kidney Transplantation (Renal Failure):** Ultimate method in the correction of acute **renal failures (kidney failure)**. A functioning kidney is transplanted from a donor, **preferably a close relative**, to minimise its chances of rejection by the host's immune system.
- **Renal Calculi:** Stone or insoluble mass of crystallised salts (**oxalates**, etc.) formed within the kidney.
- **Glomerulonephritis:** Inflammation of the glomeruli of the kidney.

8. Worked Examples & Foundational Concept Walkthroughs

The following worked examples cover core quantitative, physiological, and conceptual problem types essential for NEET UG preparation.

Worked Example 1: Quantitative Analysis of GFR, Reabsorption, and Renal Blood Flow

Problem: In a healthy adult human, calculate: (a) the percentage of glomerular filtrate that is reabsorbed by the renal tubules per day, and (b) the total volume of blood filtered by the kidneys in one hour, given that normal renal blood filtration is $1100–1200\,\text{mL/min}$.

Solution:

(a) The Glomerular Filtration Rate (GFR) in a healthy individual is $125\,\text{mL/min}$, which equals $180\,\text{L/day}$ ($125\,\text{mL/min} \times 60\,\text{min/hr} \times 24\,\text{hr/day} = 180,000\,\text{mL} = 180\,\text{L}$). The average daily urine output is $1.5\,\text{L/day}$. Therefore, the volume reabsorbed per day is $180 - 1.5 = 178.5\,\text{L}$:

$$\text{Percentage Reabsorption} = \frac{178.5\,\text{L}}{180\,\text{L}} \times 100\% \approx 99.17\%\ (\text{nearly } 99\%)$$

(b) Taking the upper standard rate of $1200\,\text{mL/min}$ (which represents $\frac{1}{5}\text{th}$ of ventricular output per minute):

$$\text{Blood filtered per hour} = 1200\,\text{mL/min} \times 60\,\text{min} = 72,000\,\text{mL} = 72\,\text{litres/hour}$$

(Using the lower bound of $1100\,\text{mL/min}$, it is $1100 \times 60 = 66\,\text{litres/hour}$, so the range is $66–72\,\text{L/hour}$).

Worked Example 2: Statement Verification (True / False Analysis)

Problem: Evaluate whether each of the following five physiological statements is **True** or **False**, and provide the scientific reason for each:

a. Micturition is carried out by a reflex.

b. ADH helps in water elimination, making the urine hypotonic.

c. Protein-free fluid is filtered from blood plasma into the Bowman's capsule.

d. Henle's loop plays an important role in concentrating the urine.

e. Glucose is actively reabsorbed in the proximal convoluted tubule.

Solution:

- **(a) True:** Stretching of the urinary bladder activates stretch receptors that signal the CNS, which sends motor commands to contract bladder smooth muscles and relax the urethral sphincter via the *micturition reflex*.
- **(b) False:** ADH (antidiuretic hormone / vasopressin) facilitates *water reabsorption* from the latter parts of the tubule, preventing diuresis and making the urine **hypertonic (concentrated)**, not hypotonic.
- **(c) True:** During ultrafiltration across the glomerular membrane, almost all constituents of blood plasma except proteins pass into the lumen of Bowman's capsule.
- **(d) True:** Henle's loop, together with the *vasa recta*, establishes the counter-current mechanism and the medullary osmolarity gradient (300 to $1200\ \mathrm{mOsmol\,L^{-1}}$) required to concentrate urine.
- **(e) True:** Essential nutrients like glucose and amino acids are actively reabsorbed in the PCT.

Worked Example 3: Evolutionary Adaptation of Terrestrial Nitrogenous Excretion

Problem: Why are terrestrial animals generally either ureotelic or uricotelic, rather than ammonotelic?

Solution: Ammonia is the most toxic form of nitrogenous waste and requires a very large volume of water for its dissolution and elimination. On land, water availability is limited compared to aquatic habitats. Therefore, terrestrial adaptation necessitated the conversion of toxic ammonia into less toxic nitrogenous wastes — **urea** (in mammals and terrestrial amphibians) or **uric acid** (in reptiles, birds, land snails, and insects, excreted as a pellet or paste with minimum water loss) — in order to **conserve water** inside the body.

Worked Example 4: Rapid Identification & Concept Completion

Problem: Identify or complete the following key excretory terms:

1. A chordate animal having flame cells (protonephridia) as excretory structures.
2. Cortical portions projecting between the medullary pyramids in the human kidney.
3. A loop of capillary running parallel to Henle's loop.
4. The ascending limb of Henle's loop is __(i)__ to water whereas the descending limb is __(ii)__ to it.
5. Dialysis fluid contains all the constituents as in plasma except __(iii)__.

Solution:

1. ***Amphioxus*** (a cephalochordate).
2. **Columns of Bertini** (renal columns).
3. ***Vasa recta***.
4. (i) **impermeable**; (ii) **permeable**.
5. (iii) **nitrogenous wastes** (such as urea).

9. NEET UG Practice Questions with Complete Worked Solutions

Note: All questions below are original practice questions designed strictly in accordance with the NEET UG examination pattern (Single-Correct MCQs, Statement-Based, Assertion-Reason, and Matrix-Match formats).

Practice Question 1 [Single-Choice MCQ • Modes of Excretion]

Which of the following statements is **incorrect** regarding the elimination of nitrogenous wastes in animals?

(1) Ammonia is readily soluble and is generally excreted across gill surfaces in bony fishes as ammonium ions.

(2) Kidneys play a primary and major role in the removal of ammonia in ammonotelic animals.

(3) In ureotelic animals, ammonia produced by metabolism is converted into urea in the liver.

(4) Some amount of urea may be retained in the kidney matrix of certain ureotelic animals to maintain a desired osmolarity.

Correct Answer: (2)

Worked Solution: Ammonia is readily soluble in water and is excreted by diffusion across body surfaces or through gill surfaces (in fishes) as ammonium ions (NH_4^+). **Kidneys do not play any significant role in its removal.** Statements (1), (3), and (4) are completely accurate.

Practice Question 2 [Matrix-Match • Animal Excretory Organs]

Match the excretory structures in **Column I** with the organisms/groups in **Column II** and select the correct option:

Column I (Excretory Structure)	Column II (Animal / Group)
(A) Protonephridia (Flame cells)	(i) Prawns (Crustaceans)
(B) Nephridia	(ii) Cockroaches
(C) Malpighian tubules	(iii) *Amphioxus* and *Planaria*
(D) Antennal glands (Green glands)	(iv) Earthworms

(1) A-(iii), B-(iv), C-(ii), D-(i)

(2) A-(iv), B-(iii), C-(ii), D-(i)

(3) A-(iii), B-(ii), C-(iv), D-(i)

(4) A-(i), B-(iv), C-(ii), D-(iii)

Correct Answer: (1)

Worked Solution: Protonephridia or flame cells occur in Platyhelminthes (*Planaria*), rotifers, some annelids, and the cephalochordate *Amphioxus* [A-(iii)]. Nephridia are the tubular excretory structures of earthworms and other annelids [B-(iv)]. Malpighian tubules occur in insects including cockroaches [C-(ii)]. Antennal glands or green glands perform excretion in crustaceans like prawns [D-(i)].

Practice Question 3 [Single-Choice MCQ • Kidney Anatomy]

Which of the following sets of structures of a nephron is situated **entirely within the cortical region** of the human kidney?

(1) Malpighian corpuscle, PCT, and Henle's loop

(2) PCT, DCT, and Vasa recta

(3) Malpighian corpuscle, PCT, and DCT

(4) Bowman's capsule, Henle's loop, and Collecting duct

Correct Answer: (3)

Worked Solution: The Malpighian corpuscle (glomerulus + Bowman's capsule), proximal convoluted tubule (PCT), and distal convoluted tubule (DCT) are situated in the cortical region of the kidney, whereas the loop of Henle and *vasa recta* dip into the medulla, and the collecting duct extends from the cortex to the inner parts of the medulla.

Practice Question 4 [Statement-Based • Cortical vs. Juxtamedullary Nephrons]

Read the following two statements carefully:

Statement I: In the majority of nephrons in the human kidney, the loop of Henle is very long and runs deep into the medulla.
Statement II: *Vasa recta* is absent or highly reduced in cortical nephrons.

In the light of the above statements, choose the correct option:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct, but Statement II is incorrect.

(4) Statement I is incorrect, but Statement II is correct.

Correct Answer: (4)

Worked Solution: In the *majority* of nephrons (which are **cortical nephrons**), the loop of Henle is *too short* and extends only very little into the medulla; only in some nephrons (juxtamedullary nephrons) is the loop of Henle very long. Thus, Statement I is incorrect. Statement II is correct because *vasa recta* is absent or highly reduced in cortical nephrons.

Practice Question 5 [Single-Choice MCQ • Ultrafiltration & JGA]

The specialized sensitive region called the Juxtaglomerular Apparatus (JGA) is formed by cellular modifications at the point of contact between:

(1) Proximal convoluted tubule and efferent arteriole

(2) Distal convoluted tubule and afferent arteriole

(3) Distal convoluted tubule and efferent arteriole

(4) Bowman's capsule and afferent arteriole

Correct Answer: (2)

Worked Solution: JGA is a special sensitive region formed by cellular modifications in the **distal convoluted tubule (DCT)** and the **afferent arteriole** at the location of their contact.

Practice Question 6 [Single-Choice MCQ • Tubular Transport Mechanisms]

During tubular reabsorption in the nephron, which of the following groups of substances is reabsorbed **actively** from the filtrate?

(1) Urea, water, and Na^{+}

(2) Glucose, amino acids, and Na^{+}

(3) Nitrogenous wastes and water in the initial segments

(4) Urea, creatinine, and ammonia

Correct Answer: (2)

Worked Solution: Substances like **glucose, amino acids, and** Na^{+} in the filtrate are reabsorbed **actively**, whereas nitrogenous wastes are absorbed by passive transport, and reabsorption of water also occurs passively in the initial segments of the nephron.

Practice Question 7 [Assertion-Reason • PCT Structure and Function]

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Nearly all essential nutrients and $70–80\%$ of electrolytes and water are reabsorbed in the Proximal Convoluted Tubule (PCT).
Reason (R): PCT is lined by simple cuboidal brush border epithelium which markedly increases the surface area for reabsorption.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution: Both Assertion and Reason are true. The PCT is lined by simple cuboidal brush border epithelium whose microvilli vastly increase the luminal surface area, directly enabling the PCT to reabsorb nearly 100% of essential nutrients and $70–80\%$ of electrolytes and water.

Practice Question 8 [Single-Choice MCQ • Loop of Henle]

As the glomerular filtrate moves down the descending limb of Henle's loop and subsequently moves upward through the ascending limb, the filtrate becomes:

(1) Diluted in the descending limb and concentrated in the ascending limb.

(2) Concentrated in the descending limb and diluted in the ascending limb.

(3) Isotonic throughout both the descending and ascending limbs.

(4) Concentrated in both the descending and ascending limbs.

Correct Answer: (2)

Worked Solution: The descending limb of Henle's loop is permeable to water but almost impermeable to electrolytes; hence, water moves out and **concentrates the filtrate** as it moves down. Conversely, the ascending limb is impermeable to water but allows active or passive transport of electrolytes (NaCl) into the medullary fluid; hence, as the concentrated filtrate passes upward, it **gets diluted**.

Practice Question 9 [Multi-Statement Evaluation • Counter-Current Mechanism]

Consider the following four statements (A–D) regarding the counter-current mechanism in the human kidney:

A. The osmolarity gradient increases from $300\ \mathrm{mOsmol\,L^{-1}}$ in the cortex to about $1200\ \mathrm{mOsmol\,L^{-1}}$ in the inner medulla.

B. The medullary concentration gradient is mainly caused by NaCl and urea.

C. NaCl transported by the ascending limb of Henle's loop is exchanged with the descending limb of *vasa recta* and returned to the interstitium by the ascending portion of *vasa recta*.

D. Small amounts of urea enter the descending limb of Henle's loop and are transported back to the interstitium by the PCT.

Which of the above statements are **correct**?

(1) A, B, and C only

(2) A, C, and D only

(3) B, C, and D only

(4) A, B, C, and D

Correct Answer: (1)

Worked Solution: Statements A, B, and C are completely accurate. Statement D is false because small amounts of urea enter the **thin segment of the ascending limb** of Henle's loop (not the descending limb) and are transported back to the medullary interstitium by the **collecting tubule / collecting duct** (not the PCT).

Practice Question 10 [Single-Choice MCQ • Hormonal Regulation]

Which of the following correctly describes the physiological effects of Angiotensin II in the Renin-Angiotensin mechanism?

(1) It causes vasodilation and inhibits aldosterone release from the adrenal medulla.

(2) It is a powerful vasoconstrictor that increases glomerular blood pressure and stimulates the adrenal cortex to release aldosterone.

(3) It is released by the atria of the heart to decrease blood pressure and GFR.

(4) It directly suppresses osmoreceptors in the hypothalamus to promote diuresis.

Correct Answer: (2)

Worked Solution: Angiotensin II is a **powerful vasoconstrictor** that increases glomerular blood pressure and thereby increases GFR. It also activates the **adrenal cortex to release Aldosterone**, which causes reabsorption of Na^{+} and water from the distal parts of the tubule.

Practice Question 11 [Assertion-Reason • ANF vs. RAAS]

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): The Atrial Natriuretic Factor (ANF) mechanism acts as a physiological check on the Renin-Angiotensin mechanism.

Reason (R): An increase in blood flow to the atria of the heart causes the release of ANF, which induces vasodilation and decreases blood pressure.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution: While the Renin-Angiotensin mechanism raises blood pressure via vasoconstriction and $\mathrm{Na}^{+}/\mathrm{H_2O}$ reabsorption, increased atrial blood flow triggers ANF release from the heart. ANF causes vasodilation and lowers blood pressure, thereby directly opposing (acting as a check on) the Renin-Angiotensin mechanism.

Practice Question 12 [Matrix-Match • Physiological Concepts & Functions]

Match the items in **Column I** with those in **Column II** and choose the correct option:

Column I	Column II
(a) Ammonotelism	(i) Birds
(b) Bowman's capsule	(ii) Water reabsorption
(c) Micturition	(iii) Bony fish
(d) Uricotelism	(iv) Urinary bladder
(e) ADH	(v) Renal tubule

(1) a-(iii), b-(v), c-(iv), d-(i), e-(ii)

(2) a-(i), b-(v), c-(iv), d-(iii), e-(ii)

(3) a-(iii), b-(iv), c-(v), d-(i), e-(ii)

(4) a-(iii), b-(v), c-(ii), d-(i), e-(iv)

Correct Answer: (1)

Worked Solution:

- (a) Ammonotelism is exhibited by bony fishes $\rightarrow$ (iii).
- (b) Bowman's capsule is the double-walled cup at the beginning of the renal tubule $\rightarrow$ (v).
- (c) Micturition is the release of urine from the urinary bladder $\rightarrow$ (iv).
- (d) Uricotelism is exhibited by birds $\rightarrow$ (i).
- (e) ADH facilitates water reabsorption from the latter parts of the tubule $\rightarrow$ (ii).

Practice Question 13 [Single-Choice MCQ • Accessory Excretory Organs]

Which of the following correctly pairs an accessory excretory structure with the specific waste substances it eliminates?

(1) Sebaceous glands — Watery fluid containing NaCl, urea, and lactic acid

(2) Sweat glands — Sterols, hydrocarbons, and waxes

(3) Liver — Bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins, and drugs via bile

(4) Lungs — Approximately $1200\ \mathrm{mL}$ of $\mathrm{CO_2}$ per minute

Correct Answer: (3)

Worked Solution: The liver secretes bile containing bilirubin, biliverdin, cholesterol, degraded steroid hormones, vitamins, and drugs, which pass out with digestive wastes. Option (1) and (2) swap sweat glands and sebaceous glands (sweat contains NaCl, urea, and lactic acid; sebum contains sterols, hydrocarbons, and waxes). Option (4) is incorrect because lungs remove approximately $200\ \mathrm{mL/minute}$ of $\mathrm{CO_2}$ (whereas $1100\text{–}1200\ \mathrm{mL/min}$ is the renal blood flow).

Practice Question 14 [Statement-Based • Hemodialysis Procedure]

Examine the following two statements regarding the clinical procedure of **hemodialysis** in uremic patients:

Statement I: Blood drained from a convenient vein is mixed with anti-heparin and pumped into the dialysing unit.
Statement II: The dialysing fluid surrounding the coiled cellophane tube has the same composition as plasma except that it lacks nitrogenous wastes.

Choose the correct option:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct, but Statement II is incorrect.

(4) Statement I is incorrect, but Statement II is correct.

Correct Answer: (4)

Worked Solution: Statement I is incorrect because during hemodialysis, blood is drained from a **convenient artery** (not a vein) after adding an anticoagulant like **heparin** (not anti-heparin), and after clearing, it is pumped back into the body through a **vein** after adding **anti-heparin**. Statement II is completely correct.

Practice Question 15 [Single-Choice MCQ • Clinical Urinalysis & Disorders]

A clinical analysis of a patient's urine sample reveals the abnormal presence of both glucose and ketone bodies. These two conditions are termed, respectively, as _____ and _____, and are diagnostic indicators of _____:

(1) Uremia, Calculi; Glomerulonephritis

(2) Glycosuria, Ketonuria; Diabetes mellitus

(3) Ketonuria, Glycosuria; Diabetes insipidus

(4) Glycosuria, Uremia; Acute renal failure

Correct Answer: (2)

Worked Solution: The presence of glucose in urine is called **Glycosuria** and the presence of ketone bodies in urine is called **Ketonuria**. Both are clinical indicators of **Diabetes mellitus**.

10. Common NEET UG Pitfalls & High-Yield Traps to Avoid

- **Pitfall 1 — Fish Excretion (Bony vs. Marine Fishes):** Do not assume all fishes are ammonotelic! **Many bony fishes** are *ammonotelic*, whereas **marine fishes** mainly excrete urea and are *ureotelic*.
- **Pitfall 2 — Role of Kidneys in Ammonotelism:** Students often assume kidneys excrete ammonia in bony fishes. Remember: ammonia diffuses across body surfaces or gill surfaces as ammonium ions, and **kidneys do not play any significant role in its removal**.
- **Pitfall 3 — Chordate with Flame Cells:** Flame cells (protonephridia) are not restricted solely to invertebrates (like *Planaria* and rotifers); they also occur in the cephalochordate ***Amphioxus***.
- **Pitfall 4 — Descending vs. Ascending Limb Permeability:** Never confuse the two limbs of Henle's loop:
  - **Descending limb:** Permeable to water, almost impermeable to electrolytes $\rightarrow$ *concentrates* filtrate.
  - **Ascending limb:** Impermeable to water, permeable to electrolytes (NaCl) $\rightarrow$ *dilutes* filtrate.
- **Pitfall 5 — Urea Entry Site in Counter-Current:** Urea does *not* enter the descending limb of Henle's loop. Small amounts of urea enter the **thin segment of the ascending limb of Henle's loop** and are returned to the medullary interstitium by the **collecting tubule**.
- **Pitfall 6 — Renin (Kidney) vs. Rennin (Stomach):** Check spelling carefully! **Renin** (single 'n') is released by the JG cells of the kidney to convert angiotensinogen to angiotensin I, whereas *rennin* (double 'n') is a proteolytic enzyme in infant gastric juice.
- **Pitfall 7 — Artery vs. Vein and Heparin vs. Anti-heparin in Hemodialysis:** Blood is taken out from a **convenient artery + heparin** $\rightarrow$ dialysed $\rightarrow$ returned through a **vein + anti-heparin**.

11. Concise Final Recap & Numerical Fact Sheet

Parameter / Physiological Quantity	Standard Value / Key Fact
Anatomical Vertebral Level of Human Kidneys	Between last thoracic ($\mathrm{T_{12}}$) and third lumbar ($\mathrm{L_3}$) vertebra
Adult Human Kidney Dimensions & Weight	Length: 10–$12\ \mathrm{cm}$ • Width: 5–$7\ \mathrm{cm}$ • Thickness: 2–$3\ \mathrm{cm}$ • Weight: 120–$170\ \mathrm{g}$
Number of Nephrons per Kidney	Nearly 1 million (10^6) per kidney
Renal Blood Filtration per Minute	1100–$1200\ \mathrm{mL/min}$ ($\approx \frac{1}{5}$th of blood pumped by each ventricle per minute)
Glomerular Filtration Rate (GFR)	$125\ \mathrm{mL/minute}$ or $180\ \mathrm{litres/day}$
Percentage of Filtrate Reabsorbed	Nearly 99% (out of $180\ \mathrm{L}$ filtrate, only 1–$1.5\ \mathrm{L}$ is excreted as urine)
PCT Reabsorption Share	Nearly 100% of essential nutrients and 70–80% of electrolytes and water
Medullary Interstitial Osmolarity Gradient	$300\ \mathrm{mOsmol\ L^{-1}}$ (cortex) to $1200\ \mathrm{mOsmol\ L^{-1}}$ (inner medulla) — $4\times$ concentration
Normal Daily Urine Output, pH, and Urea Excreted	Volume: 1–$1.5\ \mathrm{L/day}$ • $\mathrm{pH} = 6.0$ (slightly acidic) • Urea: 25–$30\ \mathrm{g/day}$
Rate of $\mathrm{CO_2}$ Removal by Lungs	Approximately $200\ \mathrm{mL/minute}$ (plus significant quantities of water daily)